

Subject Description Form

Subject Code	EEE524
Subject Title	Green Technology and Policy in Electrical Engineering
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	1. To present fundamental concepts on energy and environmental policies and the related green technology. 2. To provide knowledge on development and current status of renewable energy sources, power, hydrogen and carbon market and policies. Emphasis will be placed on technological and socio-economical points of view. 3. To provide knowledge on concept of green technology, energy management, resource planning and demand side management.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. To understand power, hydrogen and carbon markets and green technology development. b. To explain the economics and technological principle of energy and carbon markets. c. To assess and discuss the roles and objective of energy and environment policies, as well as its implementation and development. d. To compare energy efficiency in different types of fuels and hence design management strategies on resources planning and demand side management. e. To communicate logically and lucidly through discussions and presentations.
Subject Synopsis/ Indicative Syllabus	1. Introduction: Nature of different kinds of energy and power, overview of energy and carbon market, world energy consumption, carbon emission and environmental impact, energy needs of growing economy, long term energy scenario, energy supply infrastructure, carbon neutrality 2050 target. 2. Power, hydrogen and carbon market: Energy sector reform, deregulation of energy & electricity market, economics and technological principles for power, hydrogen and carbon market operations, power, hydrogen and carbon trading models: pool and bilateral contracts, market power and its mitigation, emission trading, energy risk management, financial hedging principles, electric power industry, ancillary services. 3. Energy and environment policies: Objective & roles of energy and environment policies, existing energy/renewable energy policy and environment policy in Hong Kong, China and overseas, its related laws, its implementation and its future development, comparisons of energy policies among developed countries, related policies for competition and sustainable development, the linkages between policy, planning and management. 4. Green technology and energy management: Concepts of green technology and energy management; comparisons of energy efficiency and environment impact of different types of fuels and modes of operations, introduction to integrated resource planning and portfolio, management (IRP & PM) for the right mix of generation types, transmission and conservation, demand side energy management. 5. Energy and environment policies & Green technology case studies: Sharing and discussions on the real development cases of energy and environment policies and green technology in the world, Comparing and designing the new SCA in Hong

	Kong, renewable energy policy review and market design.						
Teaching/Learning Methodology	The concept of energy policy and management will be presented through lectures and tutorials on local and international case studies. Students are expected to take initiative to learn through the process of engagement and participation in lectures and tutorial sessions. Students will be required to form groups to work through a mini-project for a selected topic. Mini-Projects are used to enhance students learning experiences and practical applications.						
	Teaching/Learning Methodology	Outcomes					
		a	b	c	d	e	
	Lectures	✓	✓	✓	✓		
	Tutorials	✓	✓	✓	✓		
	Mini-projects		✓	✓	✓	✓	
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed				
			a	b	c	d	e
	1. Examination	60%	✓	✓	✓	✓	
	2. Class Test	15%	✓	✓	✓		
	3. Essay Assignment	10%	✓	✓	✓	✓	
	4. Mini-project & report	15%		✓	✓	✓	✓
	Total	100%					
	The subject outcomes on concepts, assessment and implementations of energy policy and management are evaluated by means of examination, quizzes and tests. The outcomes on practical applications and implementations of energy policy and management, as well as technical writing, are evaluated by essay assignment, mini-project and reports.						
Student Study Effort Expected	Class contact:						
	▪ Lecture/Tutorial				39 Hrs.		
	Other student study effort:						
	▪ Mini-project discussion/Report/Essay				20 Hrs.		
	▪ Self-study				46 Hrs.		
	Total student study effort				105 Hrs.		
Reading List and References	Reference books:						
	1. D. Mulvaney, Green Technology: An A-to-Z Guide, SAGE Publications, 2011						
	2. F. Kreith and D.Y. Goswami, Energy Management and Conservation Handbook, Boca Raton: CRC Press, 2008						
	3. M. Chick, Electricity and Energy Policy in Britain, France and the United States since 1945, Cheltenham, Northampton, Mass: Edward Elgar, 2007						
	4. K. Mallon, Renewable Energy Policy and Politics: A Handbook for Decision-making, London, Sterling, VA: Eathscan, 2006						
	5. B.L. Capehart, W.C. Turner and W.J. Kennedy, Guide to Energy Management, Fairmont Press, New York: distributed by Dekker, 2003						